

Answer Key & Teacher Notes

Session 1 – Trapezoids by Decomposing • Session 2 – The Trapezoid Area Formula

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

TALK PROMPTS

- Why must the two end triangles be the same size on an isosceles tabletop?
- How does cutting a trapezoid into simpler shapes make its area easier to find?
- How do you know the leftover part of the long base splits into two equal pieces?

WATCH FOR

- Using the whole leftover (long base – short base) as the base of EACH of two triangles instead of splitting it in half.
- Finding only the rectangle and forgetting to add the triangles.
- Using a slanted side as the height of a triangle piece.

PARTNER TASK

Window Frame. A trapezoid window has bases 13 cm and 25 cm, height 11 cm, and equal slanted sides. One partner finds the rectangle (143 cm^2); the other finds both triangles (66 cm^2). Add to get 209 cm^2 , then switch jobs on the bus shelter roof.

SESSION 2 • TEACHER NOTES

TALK PROMPTS

- Why does the formula add the two bases before anything else happens?
- How is this formula like finding the average of the two bases?
- What does the sum of the two bases tell you before you multiply by the height?

WATCH FOR

- Forgetting the $\frac{1}{2}$ (finding $(b_1 + b_2) \times h$, which is the area of two trapezoids).
- Using a slanted side as the height or as a base.
- Missing base: dividing the area by the height without doubling first.

PARTNER TASK

Ramp Side. A skate-ramp side is a right trapezoid with bases 12 ft and 20 ft and height 7 ft. One partner uses the formula; the other decomposes it into a rectangle (84 ft^2) and a triangle (28 ft^2). Both should get 112 ft^2 . Explain why the methods must agree.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.1	1	1	1	2	2	2	2	3	3	1	1
S1 • Form B	6.G.A.1	1	1	1	2	2	2	2	3	3	1	1
S2 • Form A	6.G.A.1	1	2	1	1	2	2	3	2	3	1	1
S2 • Form B	6.G.A.1	1	2	1	1	2	2	3	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

trapezoid: A **quadrilateral** with one pair of parallel sides (the bases).

height: The **perpendicular** distance between the two bases.

decompose: To **break** a shape into smaller shapes you know.

- Cut a trapezoid into a **rectangle** and one or two triangles.
- Find the area of each piece, then **add** the areas.
- Together, the triangle bases equal the **difference** of the two bases.

Example: Rectangle: $6 \times 5 = 30 \text{ cm}^2$. Leftover: $14 - 6 = 8$, so each triangle base is **4 cm**. Both triangles: $2 \times (\frac{1}{2} \times 4 \times 5) = 20 \text{ cm}^2$. Answer: Area = $30 + 20 = 50 \text{ cm}^2$

YOU TRY

1. Rectangle $6 \times 4 = 24 \text{ in}^2$; triangle $\frac{1}{2} \times 4 \times 4 = 8 \text{ in}^2$; total 32 in^2
2. a. $(14 - 8) \div 2 = 3 \text{ m}$ b. $(20 - 12) \div 2 = 4 \text{ cm}$

EXIT TICKET

1. Rectangle $9 \times 12 = 108 \text{ yd}^2$; triangle $\frac{1}{2} \times 8 \times 12 = 48 \text{ yd}^2$; total 156 yd^2
2. a. $12 \times 7 = 84 \text{ cm}^2$ b. $2 \times (\frac{1}{2} \times 4 \times 7) = 28 \text{ cm}^2$
3. **A** (Add the areas of the pieces.)

PRACTICE FORM A

1. a. $(22 - 10) \div 2 = 6 \text{ in}$ b. $(18 - 8) \div 2 = 5 \text{ m}$ c. $(20 - 12) \div 2 = 4 \text{ cm}$ d. $(27 - 17) \div 2 = 5 \text{ m}$
2. a. $10 \times 15 = 150 \text{ in}^2$ b. $(22 - 10) \div 2 = 6 \text{ in each}$; $2 \times (\frac{1}{2} \times 6 \times 15) = 90 \text{ in}^2$ c. $150 + 90 = 240 \text{ in}^2$
3. Rectangle $16 \times 10 = 160 \text{ ft}^2$; triangle base $24 - 16 = 8 \text{ ft}$, $\frac{1}{2} \times 8 \times 10 = 40 \text{ ft}^2$; total 200 ft^2
4. Rectangle $8 \times 6 = 48 \text{ m}^2$; each triangle base 5 m , both triangles 30 m^2 ; total 78 m^2
5. Rectangle $11 \times 13 = 143 \text{ yd}^2$; triangle $\frac{1}{2} \times 8 \times 13 = 52 \text{ yd}^2$; total 195 yd^2
6. Tray: $19 \times 16 = 304$; triangles $2 \times (\frac{1}{2} \times 5 \times 16) = 80$; tray 384 in^2 . Mat $24 \times 16 = 384 \text{ in}^2$. Yes, same area.
7. Rectangle $17 \times 10 = 170 \text{ m}^2$; each triangle base 5 m ; triangles $2 \times (\frac{1}{2} \times 5 \times 10) = 50 \text{ m}^2$; total 220 m^2
8. a. rectangle $10 \times 6 = 60 \text{ ft}^2$; triangles 24 ft^2 (two of $\frac{1}{2} \times 4 \times 6$, or one of $\frac{1}{2} \times 8 \times 6$) b. 84 ft^2 c. $84 \times 8 = \$672$
9. Kwame found only the rectangle and left out the two triangles. Each triangle base is $(21 - 11) \div 2 = 5 \text{ m}$, so the triangles add $2 \times (\frac{1}{2} \times 5 \times 10) = 50 \text{ m}^2$. Area = $110 + 50 = 160 \text{ m}^2$.
10. **B** (one or two triangles)
11. **C** (46 cm^2)

Self-check word COMBINE

PRACTICE FORM B

1. a. $(20 - 12) \div 2 = 4 \text{ in}$ b. $(16 - 6) \div 2 = 5 \text{ m}$ c. $(21 - 9) \div 2 = 6 \text{ cm}$ d. $(23 - 15) \div 2 = 4 \text{ ft}$
2. a. $12 \times 9 = 108 \text{ in}^2$ b. $(20 - 12) \div 2 = 4 \text{ in each}$; $2 \times (\frac{1}{2} \times 4 \times 9) = 36 \text{ in}^2$ c. $108 + 36 = 144 \text{ in}^2$
3. Rectangle $14 \times 9 = 126 \text{ ft}^2$; triangle base $22 - 14 = 8 \text{ ft}$, $\frac{1}{2} \times 8 \times 9 = 36 \text{ ft}^2$; total 162 ft^2
4. Rectangle $6 \times 8 = 48 \text{ m}^2$; each triangle base 5 m , both triangles 40 m^2 ; total 88 m^2
5. Rectangle $9 \times 14 = 126 \text{ yd}^2$; triangle $\frac{1}{2} \times 6 \times 14 = 42 \text{ yd}^2$; total 168 yd^2
6. Tray: $14 \times 12 = 168$; triangles $2 \times (\frac{1}{2} \times 5 \times 12) = 60$; tray 228 in^2 . Mat $19 \times 12 = 228 \text{ in}^2$. Yes, same area.
7. Rectangle $15 \times 8 = 120 \text{ m}^2$; each triangle base 5 m ; triangles $2 \times (\frac{1}{2} \times 5 \times 8) = 40 \text{ m}^2$; total 160 m^2
8. a. rectangle $12 \times 5 = 60 \text{ ft}^2$; triangles 20 ft^2 (two of $\frac{1}{2} \times 4 \times 5$, or one of $\frac{1}{2} \times 8 \times 5$) b. 80 ft^2 c. $80 \times 9 = \$720$
9. Amina found only the rectangle and left out the two triangles. Each triangle base is $(23 - 13) \div 2 = 5 \text{ m}$, so the triangles add $2 \times (\frac{1}{2} \times 5 \times 8) = 40 \text{ m}^2$. Area = $104 + 40 = 144 \text{ m}^2$.
10. **C** (height)
11. **C** (52 cm^2)

Self-check word SECTION

GUIDED NOTES

base: One of the two **parallel** sides of a trapezoid.

height: The **perpendicular** distance between the two bases.

formula: A **rule** written with letters and numbers.

right trapezoid: A trapezoid with two **right** angles.

- Area of a trapezoid: $A = \frac{1}{2} \times (b_1 + b_2) \times h$.
- **Add** the bases first, then multiply by the height, then take half.
- Two copies of a trapezoid make a **parallelogram**, so the trapezoid is half of it.

Example: Add the bases: $6 + 12 = 18$. Multiply by the height: $18 \times 4 = 72$. Take half: $72 \div 2 = 36$. Answer: $A = 36 \text{ m}^2$

YOU TRY

- $\frac{1}{2} \times (5 + 9) \times 8 = \frac{1}{2} \times 14 \times 8 = 56 \text{ cm}^2$
- a. $\frac{1}{2} \times 10 \times 4 = 20 \text{ m}^2$ b. $\frac{1}{2} \times 22 \times 5 = 55 \text{ in}^2$

EXIT TICKET

- a. $\frac{1}{2} \times 14 \times 4 = 28 \text{ cm}^2$ b. $\frac{1}{2} \times 26 \times 3 = 39 \text{ ft}^2$
- $\frac{1}{2} \times (12 + 20) \times 6 = \frac{1}{2} \times 32 \times 6 = 96 \text{ ft}^2$
- C** ($\frac{1}{2} \times (5 + 13) \times 4$)

PRACTICE FORM A

- a. $\frac{1}{2} \times 24 \times 8 = 96 \text{ cm}^2$ b. $\frac{1}{2} \times 48 \times 15 = 360 \text{ in}^2$ c. $\frac{1}{2} \times 30 \times 4 = 60 \text{ ft}^2$ d. $\frac{1}{2} \times 16 \times 5 = 40 \text{ m}^2$
- a. $14 + 26 = 40 \text{ mi}$ b. $\frac{1}{2} \times 40 \times 10 = 200 \text{ mi}^2$ c. 40 mi
- $\frac{1}{2} \times (18 + 30) \times 11 = \frac{1}{2} \times 48 \times 11 = 264 \text{ in}^2$
- $\frac{1}{2} \times (8 + 20) \times 7 = 98 \text{ m}^2$
- No. $\frac{1}{2} \times (11 + 17) \times 20 = 280 \text{ cm}^2$. Talia forgot to take half.
- $\frac{1}{2} \times (16 + 22) \times 10 = 190 \text{ ft}^2$; $190 \times 2 = \$380$
- M: $\frac{1}{2} \times 36 \times 8 = 144 \text{ cm}^2$; N: $\frac{1}{2} \times 36 \times 8 = 144 \text{ cm}^2$. Same area, because the bases have the same sum (36) and the heights match.
- a. $2 \times 84 \div 7 = 24 \text{ m}$ b. $24 - 10 = 14 \text{ m}$
- Sofia used only one base. The formula adds both bases first: $A = \frac{1}{2} \times (7 + 21) \times 5 = \frac{1}{2} \times 28 \times 5 = 70 \text{ m}^2$.
- B** (48 cm^2)
- C** (add the bases)

Self-check word FORMULA

PRACTICE FORM B

- a. $\frac{1}{2} \times 20 \times 6 = 60 \text{ cm}^2$ b. $\frac{1}{2} \times 36 \times 10 = 180 \text{ in}^2$ c. $\frac{1}{2} \times 44 \times 5 = 110 \text{ ft}^2$ d. $\frac{1}{2} \times 16 \times 9 = 72 \text{ m}^2$
- a. $18 + 30 = 48 \text{ mi}$ b. $\frac{1}{2} \times 48 \times 12 = 288 \text{ mi}^2$ c. 48 mi
- $\frac{1}{2} \times (16 + 24) \times 13 = \frac{1}{2} \times 40 \times 13 = 260 \text{ in}^2$
- $\frac{1}{2} \times (6 + 18) \times 9 = 108 \text{ m}^2$
- No. $\frac{1}{2} \times (13 + 19) \times 15 = 240 \text{ cm}^2$. Marcus forgot to take half.
- $\frac{1}{2} \times (18 + 26) \times 10 = 220 \text{ ft}^2$; $220 \times 3 = \$660$
- M: $\frac{1}{2} \times 36 \times 7 = 126 \text{ cm}^2$; N: $\frac{1}{2} \times 36 \times 7 = 126 \text{ cm}^2$. Same area, because the bases have the same sum (36) and the heights match.
- a. $2 \times 90 \div 6 = 30 \text{ m}$ b. $30 - 11 = 19 \text{ m}$
- Andre used only one base. The formula adds both bases first: $A = \frac{1}{2} \times (9 + 15) \times 6 = \frac{1}{2} \times 24 \times 6 = 72 \text{ m}^2$.
- D** (56 in^2)
- A** (two bases)

Self-check word AVERAGE

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · TRAPEZOIDS BY DECOMPOSING

1. a. $6 \times 4 = 24 \text{ cm}^2$ b. triangle base $(10 - 6) \div 2 = 2$ cm; $\frac{1}{2} \times 2 \times 4 = 4 \text{ cm}^2$ Area: $24 + 4 + 4 = 32 \text{ cm}^2$
2. Rectangle $7 \times 6 = 42$; triangle base $12 - 7 = 5$, $\frac{1}{2} \times 5 \times 6 = 15$; total 57 in^2
3. Rectangle $3 \times 4 = 12$; triangle base $8 - 3 = 5$, $\frac{1}{2} \times 5 \times 4 = 10$; total 22 ft^2
4. Rectangle $5 \times 3 = 15$; each triangle base $(11 - 5) \div 2 = 3$, $\frac{1}{2} \times 3 \times 3 = 4.5$; total $15 + 4.5 + 4.5 = 24 \text{ ft}^2$
5. The two triangle bases together equal $15 - 9 = 6$ cm, so each is 3 cm. Rectangle $9 \times 4 = 36$; triangles $2 \times (\frac{1}{2} \times 3 \times 4) = 12$; area = 48 cm^2 .

SHEET 2 · THE TRAPEZOID AREA FORMULA

1. $\frac{1}{2} \times (8 + 14) \times 5 = 55 \text{ m}^2$
2. a. $\frac{1}{2} \times 14 \times 6 = 42 \text{ ft}^2$ b. $\frac{1}{2} \times 14 \times 8 = 56 \text{ in}^2$ c. $\frac{1}{2} \times 18 \times 3 = 27 \text{ m}^2$ d. $\frac{1}{2} \times 10 \times 9 = 45 \text{ cm}^2$
3. $\frac{1}{2} \times (16 + 24) \times 10 = 200 \text{ ft}^2$; $200 \div 25 = 8$ bags
4. $2 \times 60 = 120$; $120 \div 6 = 20 = 8 + b$, so $b = 12$ in
5. **B** (40 cm^2)
6. **C** ($\frac{1}{2} \times (3 + 9) \times 4$)